

Ch. 14 — P02 (K) Q-Learning with the SimDes Gymnasium Environment

Problem Bank | Chapter 14 | Type: K | Difficulty:

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Background

The `simdes` package provides `simdes.envs.ClinicEnv`, a Gymnasium-compatible environment wrapping the SimPy clinic model. Q-learning is a model-free, off-policy RL algorithm that estimates the optimal action-value function:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right]$$

Setup

The observation is the queue length at decision epochs; the action is the number of nurses to deploy.

Tasks

(a) Inspect the environment:

What are the shapes and ranges of the observation and action spaces?

(b) Implement a tabular ε -greedy Q-learning agent. Use a Q -table of shape `(n_states, n_actions)` initialized to zero.

Fill in the two # TODO lines.

(c) Train the agent for 5000 episodes. Plot the *smoothed* episode return (rolling mean over 200 episodes). Does the agent converge? At what episode does the rolling mean stabilize?

(d) Extract the learned policy $\pi(n) = \arg \max_a Q(n, a)$ for queue lengths $n = 0, 1, \dots, 15$. Plot the policy as a step function of queue length. Does the agent deploy more nurses as the queue grows?

(e) Compare the learned policy against two heuristics over 100 evaluation episodes (no exploration, fixed seed range 10001–10100):

- Heuristic 1: always deploy 1 nurse
- Heuristic 2: always deploy 2 nurses
- Q-policy: use the learned Q -table

Report mean $\pm$ std of total return per episode for each. Is the Q-policy statistically better than Heuristic 2 at $\alpha = 0.05$?

(f) (*Bonus*) Modify the environment to simulate a **time-of-day arrival rate** that is $2\times$ higher from hours 8–12:

Re-train and report whether the learned policy adapts to peak hours.

Notes

- The state must be discretized to an integer index for tabular Q-learning. Use `int(obs)` directly since the observation is already the queue length.
- Clip observations to $[0, n_{\max}]$ to avoid index-out-of-bounds errors.
- Use `env.reset(seed=ep)` for reproducible training; use fixed seeds for evaluation.